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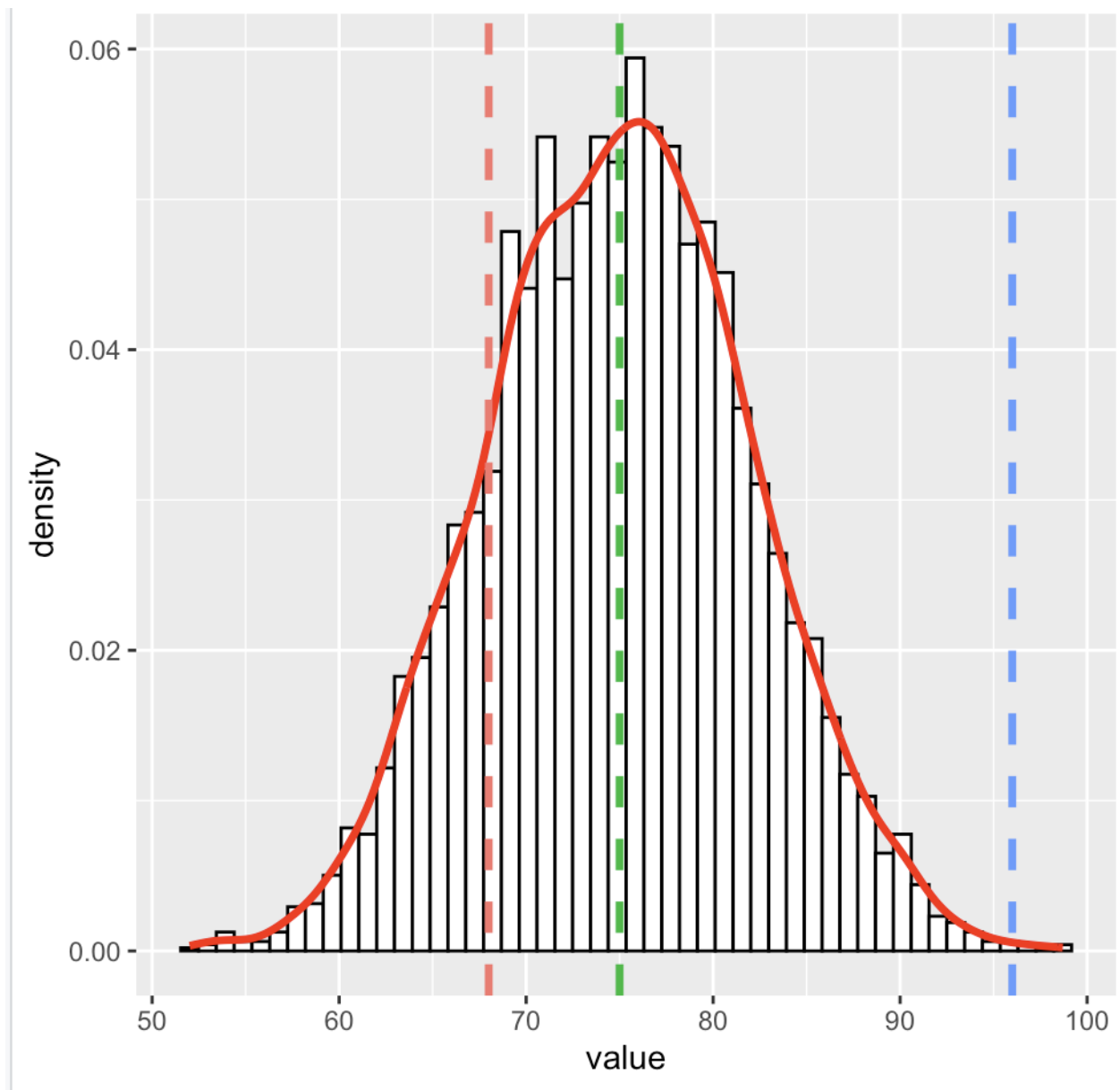
ENBC 332

Homework 3 Discussion and Outputs

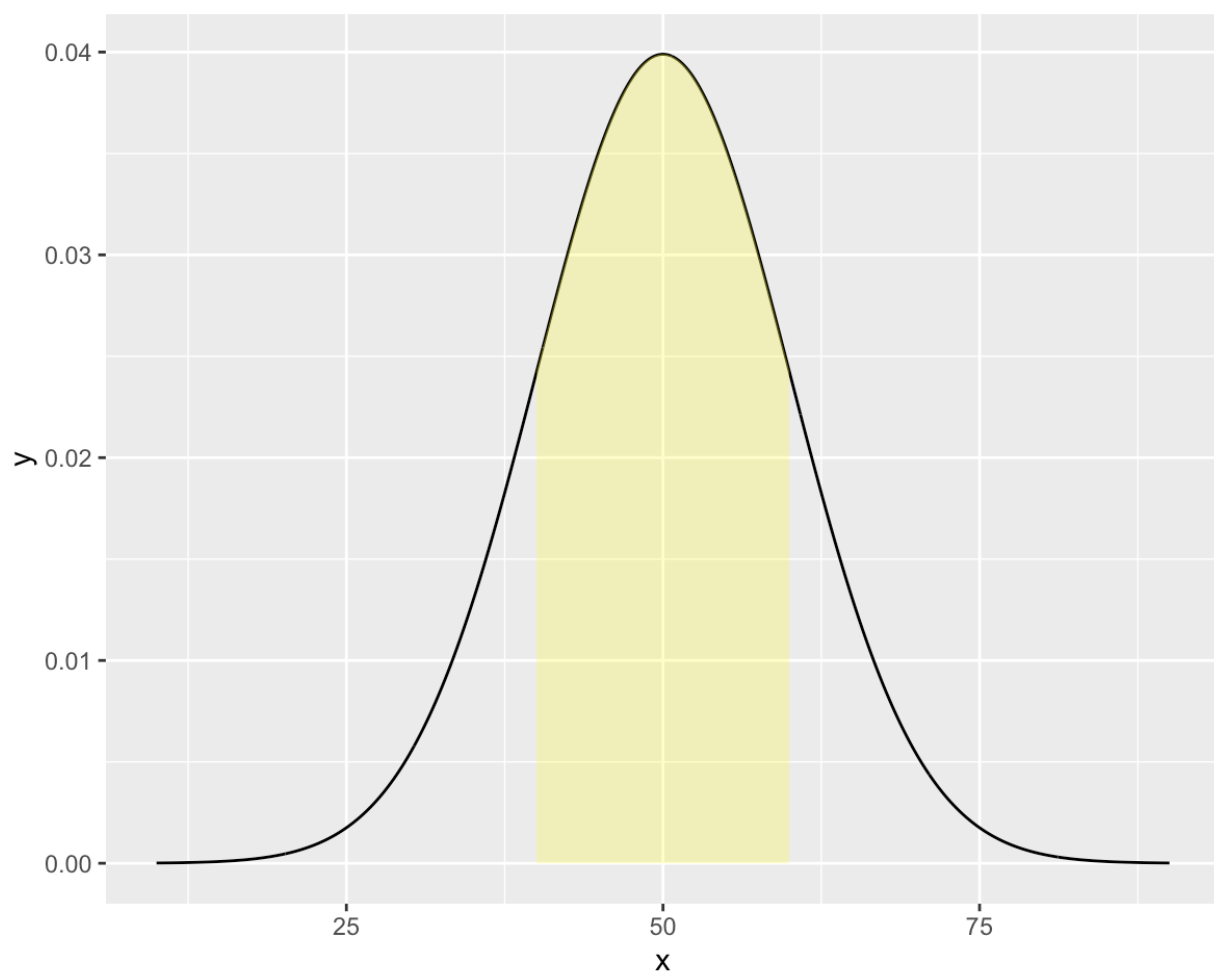
Problem 1 Part C:

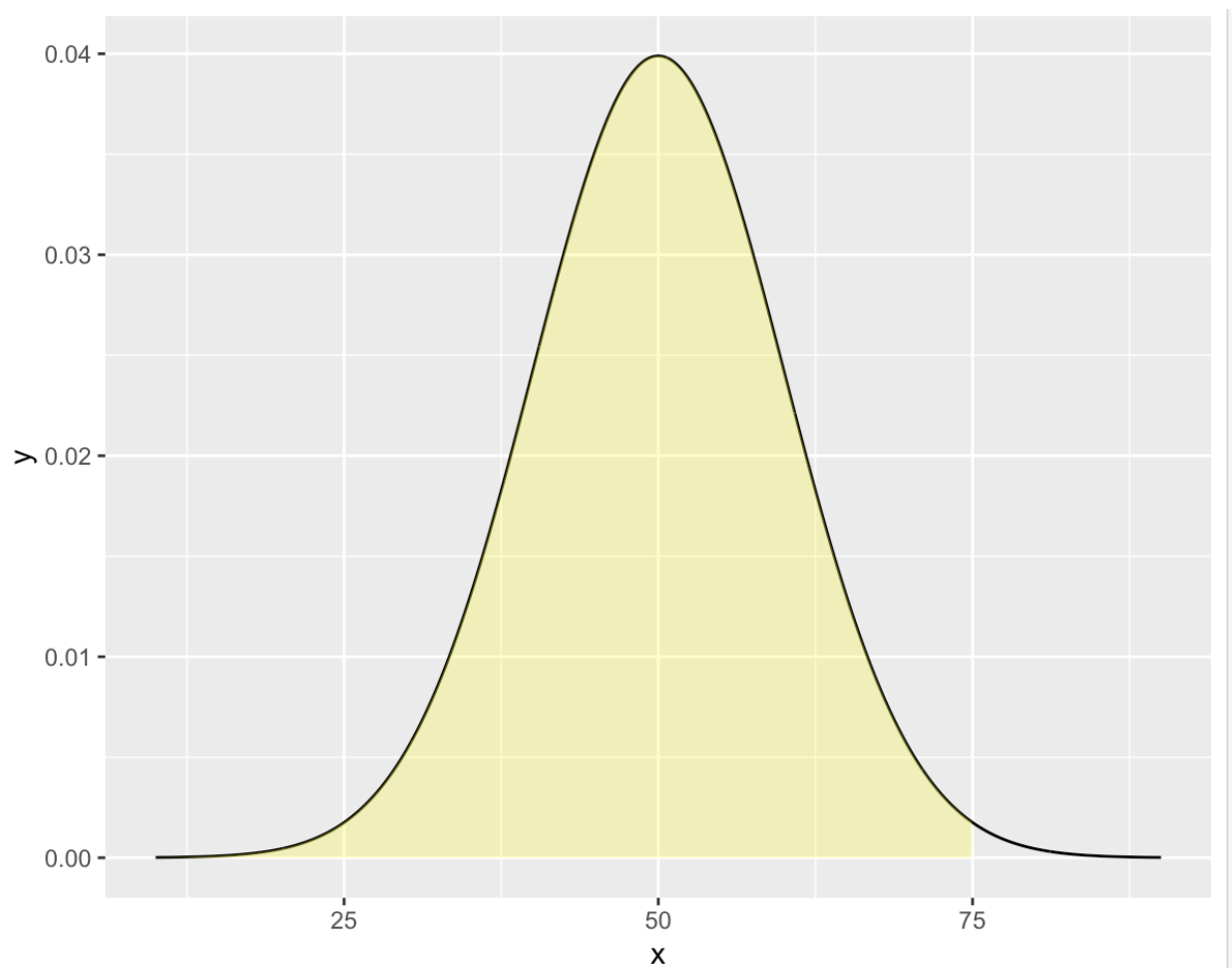
In the context of the normal distribution, the following z scores represents the number of standard deviations from. More specifically, the first z score at 68: -0.9935638 , is approximately -1 which represents one standard deviation below the average value of the dataset. For the second z score at 75, the value is extremely close to zero and represents the average of the dataset itself. For the third z score at 96, it is approximately positive 3 which represents three standard deviations above the average.

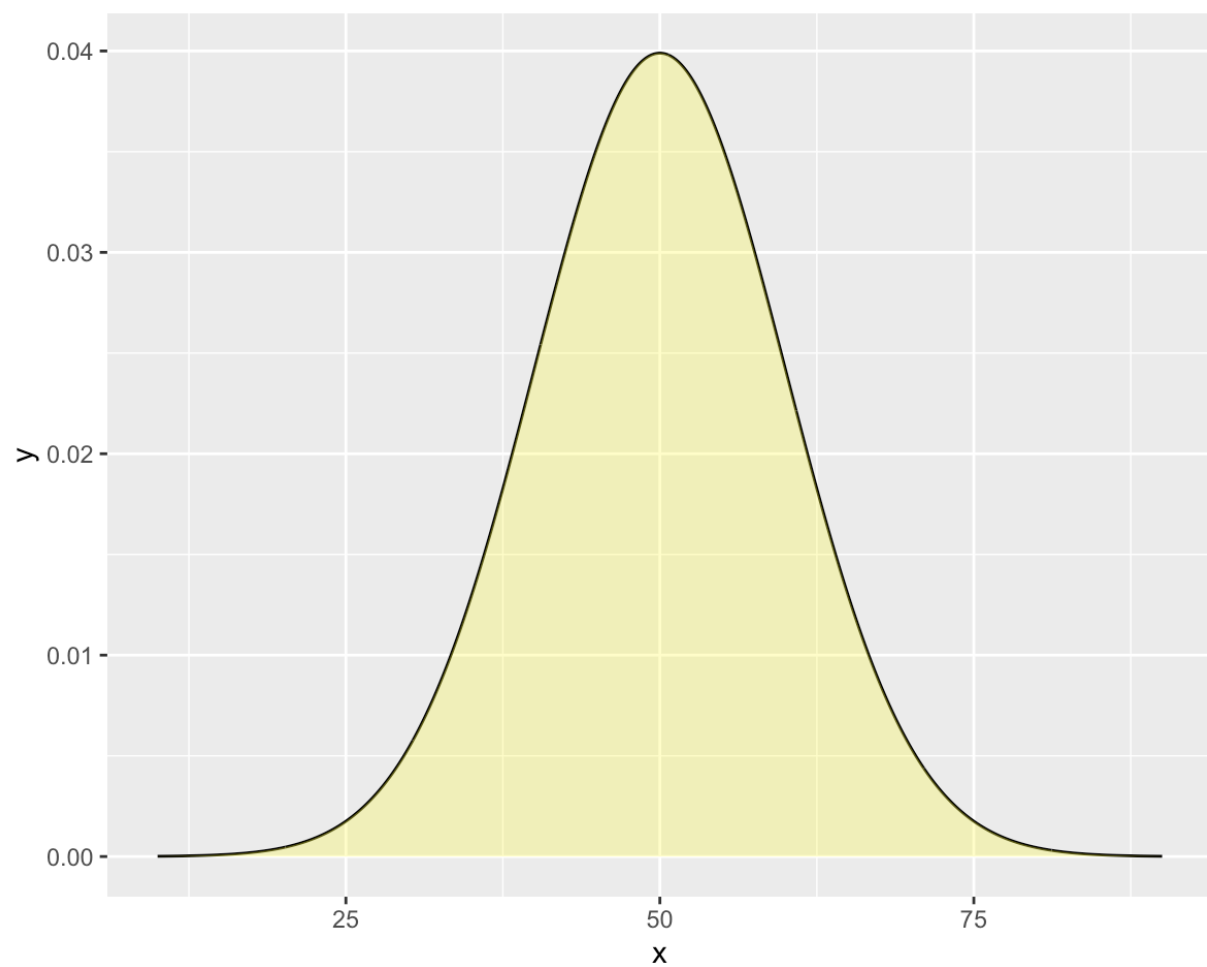
Problem 1 Plot:

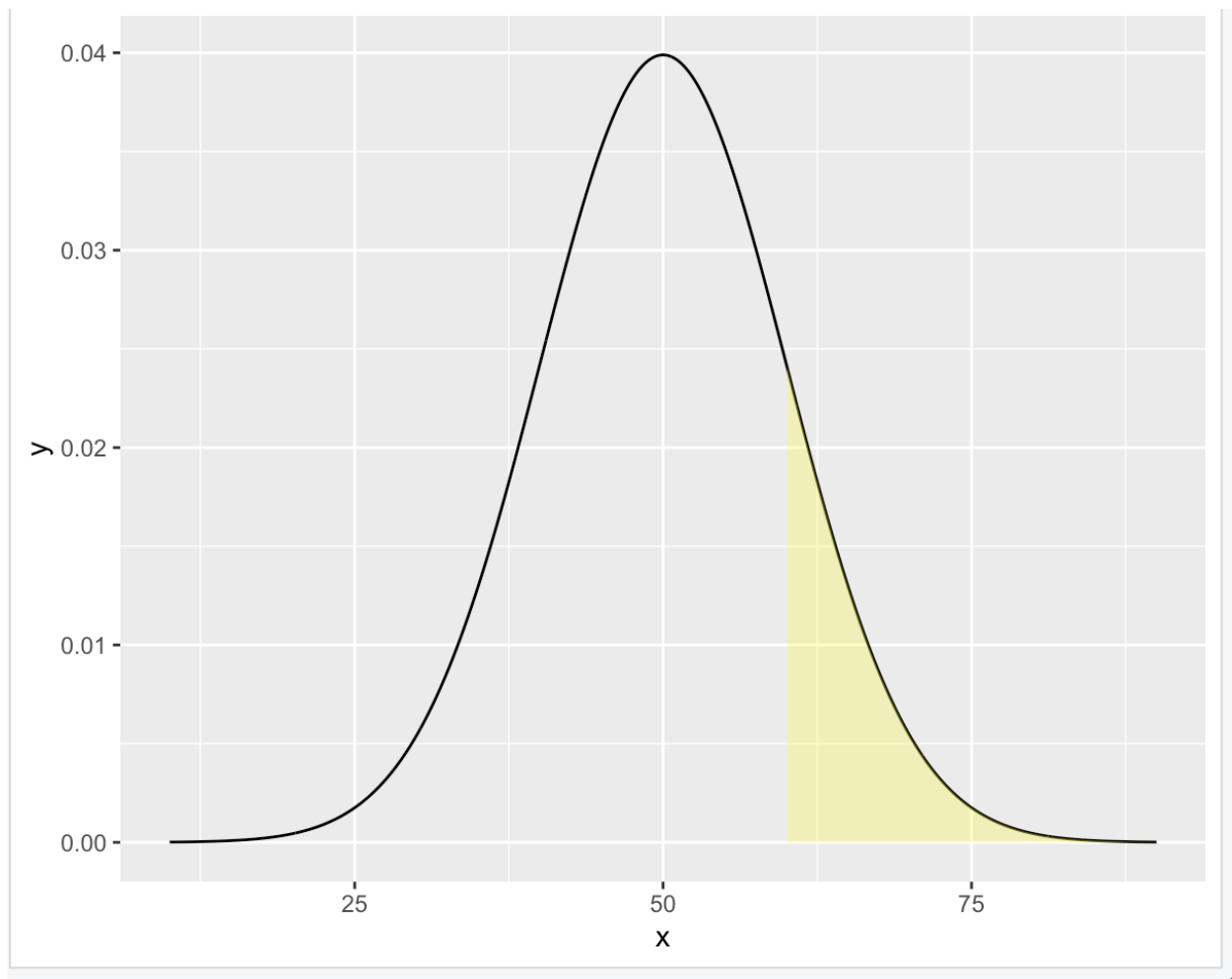


Problem 2 Plots:

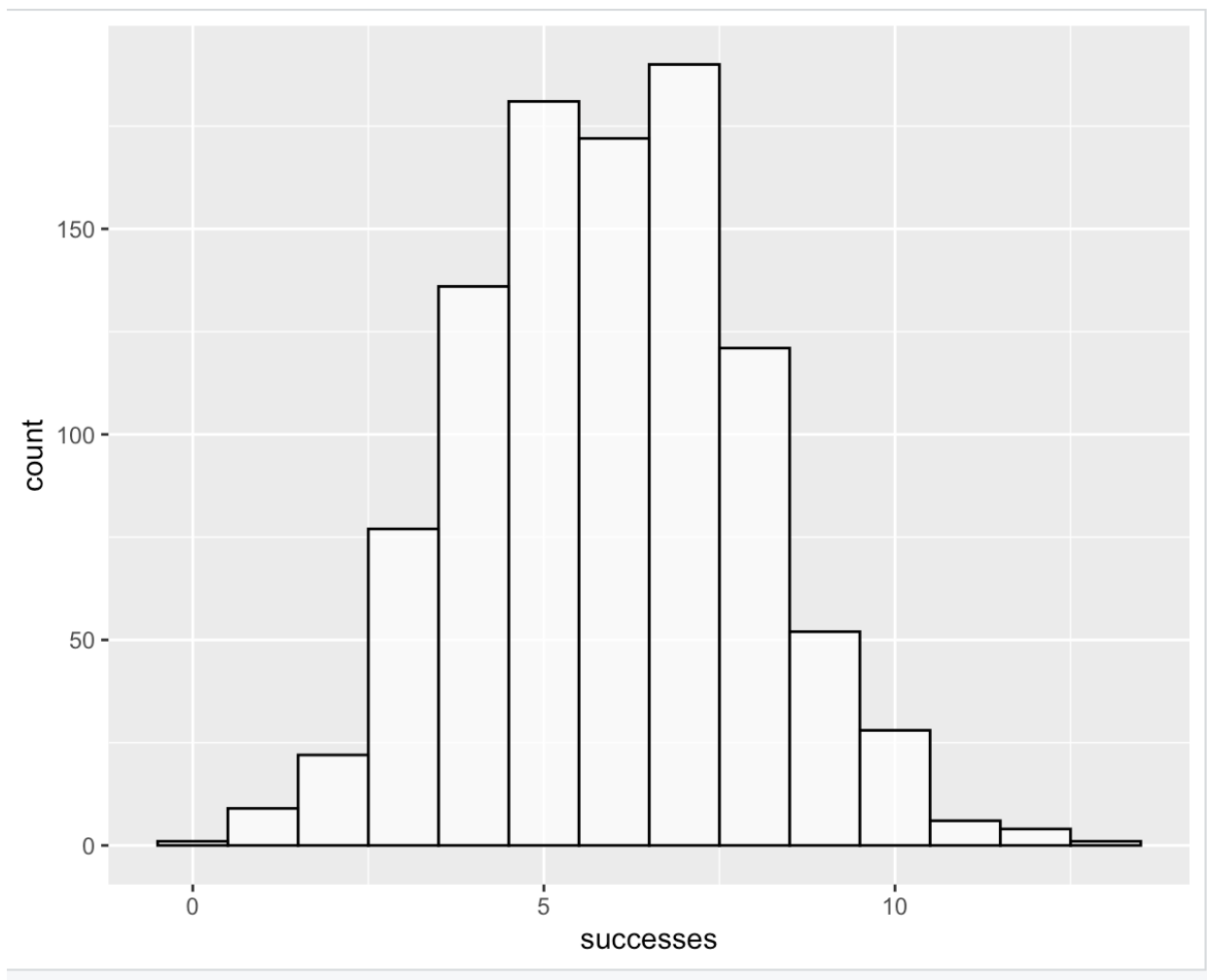


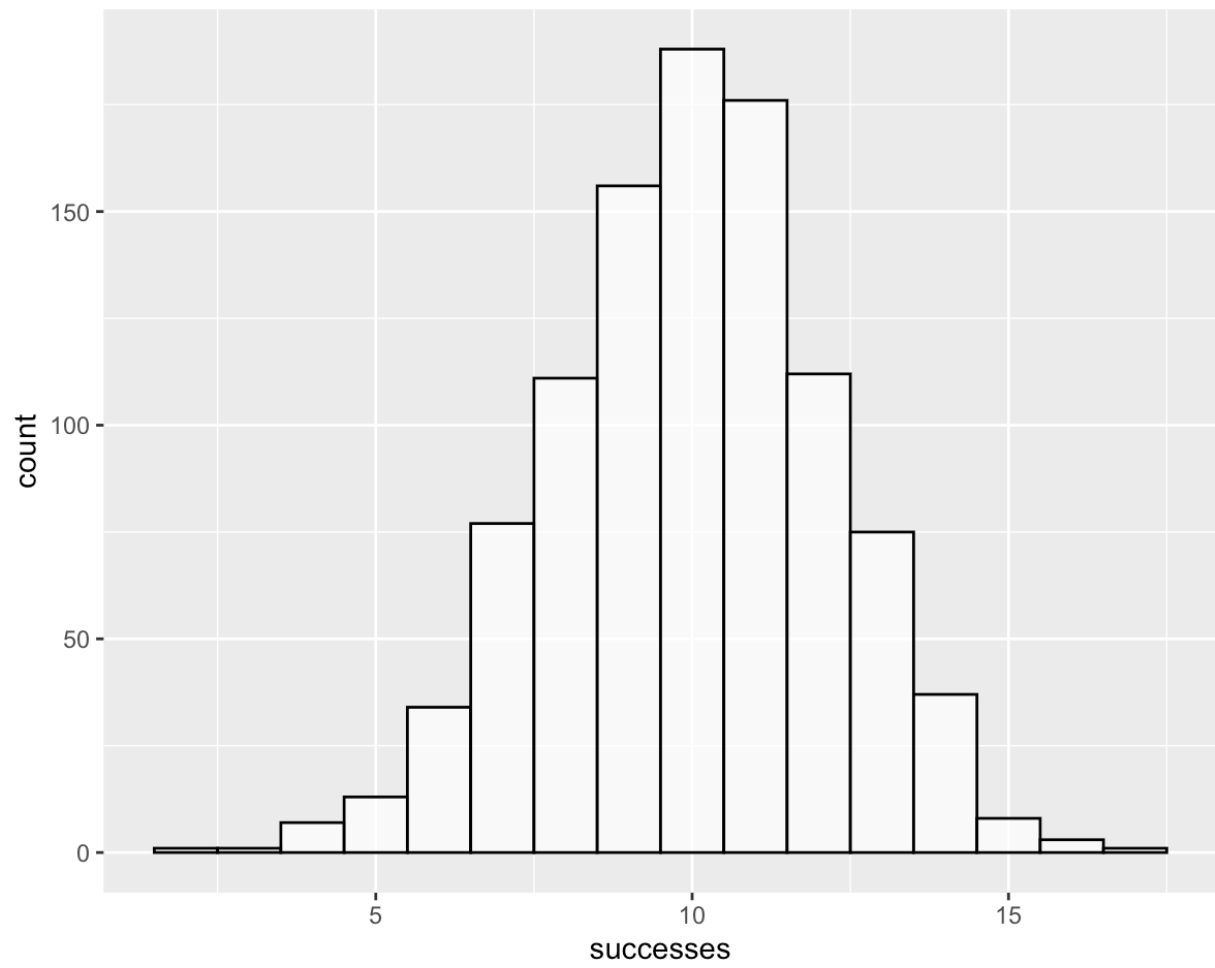


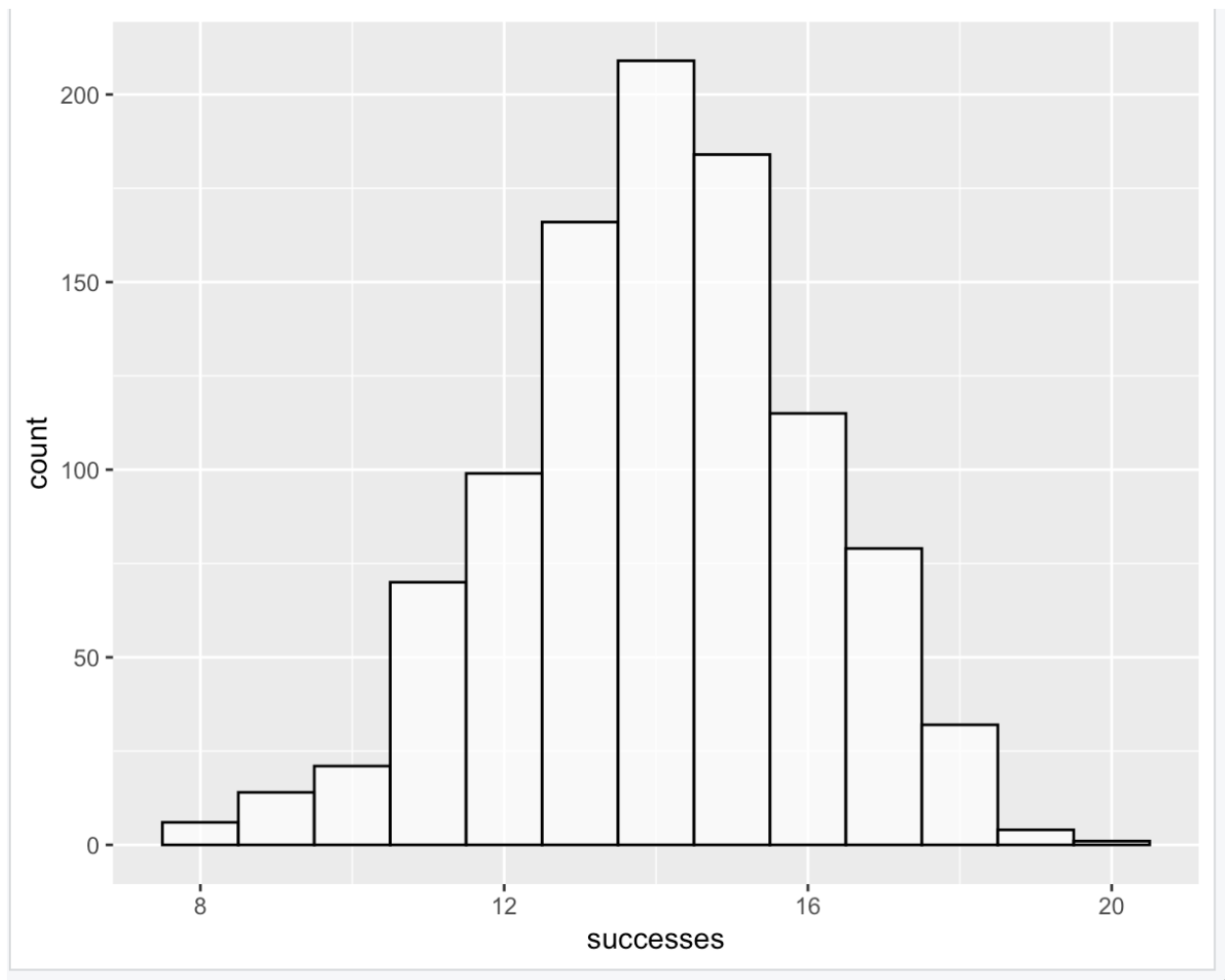




Problem 3 Plots:







Outputs:

Problem 1:

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> data <- rnorm(5000, mean = 75, sd = 7)
> df <- data.frame(value = data)
> first_plot <- ggplot(df, aes(x = value)) +
+   geom_histogram(aes(y = ..density..), bins = 50, fill = "white", color = "black") +
+   geom_density(color = "red", size = 1.2)
> # Part B
> mean_data <- mean(data)
> sd_data <- sd(data)
> z_score1 <- (68 - mean_data) / sd_data
> z_score2 <- (75 - mean_data) / sd_data
> z_score3 <- (96 - mean_data) / sd_data
> cat("Z Score at 68:", z_score1)
Z Score at 68: -0.9935638
> cat("Z Score at 75:", z_score2)
Z Score at 75: 0.004873002
> cat("Z Score at 96:", z_score3)
Z Score at 96: 3.000184
> data_points <- c(68, 75, 96)
> vertical_lines <- data.frame(
+   x_intercept = data_points,
+   label = paste("Value:", data_points)
+ )
> final_plot <- first_plot +
+   geom_vline(
+     data = vertical_lines,
+     aes(xintercept = x_intercept, color = label), # Map color to the label
+     size = 1.2,
+     linetype = "dashed",
+   )
> print(final_plot)
>

```

Problem 2:

```

> # Variables
> mean_val <- 50
> sd_val <- 10
> # Plot
> x <- seq(mean_val - 4*sd_val, mean_val + 4*sd_val, length.out = 1000)
> y <- dnorm(x, mean = mean_val, sd = sd_val)
> df <- data.frame(x, y)
> # Part A
> # Calculations for probability between 40 and 60
> prob_a <- pnorm(60, mean = mean_val, sd = sd_val) - pnorm(40, mean = mean_val, sd = sd_val)
> cat("Probability between 40 and 60:", prob_a)
Probability between 40 and 60: 0.6826895
> # Generate plot that highlights the area between 40 and 60
> ggplot(df, aes(x, y)) +
+   geom_line() +
+   geom_ribbon(data = subset(df, x > 40 & x < 60), aes(ymin = 0, ymax = y), fill = "yellow", alpha = 0.3)
> # Part B
> # Calculations for probability between 5 and 75
> prob_b <- pnorm(75, mean = mean_val, sd = sd_val) - pnorm(5, mean = mean_val, sd = sd_val)
> cat("Probability between 5 and 75:", prob_b)
Probability between 5 and 75: 0.9937869
> # Generate plot that highlights the area between 5 and 75
> ggplot(df, aes(x, y)) +
+   geom_line() +
+   geom_ribbon(data = subset(df, x > 5 & x < 75), aes(ymin = 0, ymax = y), fill = "yellow", alpha = 0.3)
> # Part C
> # Calculations for probability greater than 60
> prob_c <- 1 - pnorm(60, mean = mean_val, sd = sd_val)
> cat("Probability greater than 60:", prob_c)
Probability greater than 60: 0.1586553
> # Generate plot that highlights the area greater than 60
> ggplot(df, aes(x, y)) +
+   geom_line() +
+   geom_ribbon(data = subset(df, x > 60), aes(ymin = 0, ymax = y), fill = "yellow", alpha = 0.3)
> # Part D
> # Calculations for probability less than 90
> prob_d <- pnorm(90, mean = mean_val, sd = sd_val)
> cat("Probability less than 90:", prob_d)
Probability less than 90: 0.9999683
> # Generate plot that highlights the area less than 90
> ggplot(df, aes(x, y)) +
+   geom_line() +
+   geom_ribbon(data = subset(df, x < 90), aes(ymin = 0, ymax = y), fill = "yellow", alpha = 0.3)

```

Problem 3:

```

> # Variables
> prob_of_success <- 0.7
> trials <- 20
> # Part A
> # Probability calculations for exactly 15 patients respond positively
> prob_a <- dbinom(15, size = trials, prob = prob_of_success)
> cat("Probability exactly 15 patients respond positively:", prob_a)
Probability exactly 15 patients respond positively: 0.1788631
> # Part B
> # Probability calculations for more than 12 patients respond positively
> prob_b <- dbinom(13, size = trials, prob = prob_of_success) +
+   dbinom(14, size = trials, prob = prob_of_success) +
+   dbinom(15, size = trials, prob = prob_of_success) +
+   dbinom(16, size = trials, prob = prob_of_success) +
+   dbinom(17, size = trials, prob = prob_of_success) +
+   dbinom(18, size = trials, prob = prob_of_success) +
+   dbinom(19, size = trials, prob = prob_of_success) +
+   dbinom(20, size = trials, prob = prob_of_success) +
+ cat("Probability more than 12 patients respond positively:", prob_b)
Probability more than 12 patients respond positively: 0.9937869
> # Part C
> # Probability calculations between 10 to 18 (inclusive) patients respond positively
> prob_c <- dbinom(10, size = trials, prob = prob_of_success) +
+   dbinom(11, size = trials, prob = prob_of_success) +
+   dbinom(12, size = trials, prob = prob_of_success) +
+   dbinom(13, size = trials, prob = prob_of_success) +
+   dbinom(14, size = trials, prob = prob_of_success) +
+   dbinom(15, size = trials, prob = prob_of_success) +
+   dbinom(16, size = trials, prob = prob_of_success) +
+   dbinom(17, size = trials, prob = prob_of_success) +
+   dbinom(18, size = trials, prob = prob_of_success)
> cat("Probability between 10 to 18 (inclusive) patients respond positively:", prob_c)
Probability between 10 to 18 (inclusive) patients respond positively: 0.9752179
> # Part D
> # Plot for 30 percent probability of success
> data1 <- rbinom(1000, size = trials, prob = 0.3)
> df1 <- data.frame(successes = data1)
> ggplot(df1, aes(x = successes)) +
+   geom_histogram(binwidth = 1, fill = "white", color = "black", alpha = 0.7)
> # Plot for 50 percent probability of success
> data2 <- rbinom(1000, size = trials, prob = 0.5)
> df2 <- data.frame(successes = data2)
> ggplot(df2, aes(x = successes)) +
+   geom_histogram(binwidth = 1, fill = "white", color = "black", alpha = 0.7)
> # Plot for 70 percent probability of success
> data3 <- rbinom(1000, size = trials, prob = 0.7)
> df3 <- data.frame(successes = data3)
> ggplot(df3, aes(x = successes)) +
+   geom_histogram(binwidth = 1, fill = "white", color = "black", alpha = 0.7)

```
